Erdős Problem 700(i): An Exact Finite Feasibility Criterion via a Synchronized Carry Compiler

Nanocodex Erdős 700 project

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Abstract

For

$$f(n) = \min_{2 \le k \le \lfloor n/2 \rfloor} \gcd\left(n, \binom{n}{k}\right),$$

we give an exact finite feasibility criterion for the equality $f(n) = n/q$ for every proper prime-power component q of n. The initial reduction identifies divisibility-minimal carry obstructions. The main structural step then compiles simultaneous realization in all relevant prime bases into one compact finite integer/Boolean system of selectors, prefix products, digits, and borrows. The system has no separate variable or disjunction for each possible row, but it retains one symbolic shared multiplier and asks whether the resulting system is feasible. Taking q to be the largest prime divisor gives the maintained formulation of Erdős Problem 700(i). Taking q to be the greatest exact prime-power component gives the literal formulation in the 1978 Erdős–Szekeres paper, with the necessary prime-power exception. The reductions and both directions of the compiler are checked in Lean; they prove the encoded iff criteria. They do not yet provide the stricter direct classification readable from the factorization without solving a bespoke feasibility problem.

1 Definitions and statement

Call k *admissible for* n when $2 \le k \le \lfloor n/2 \rfloor$. For a prime p, let $C_p(n,k)$ be the number of carries in the base-p addition $k + (n-k) = n$. By Kummer's theorem,

$$C_p(n,k) = v_p\left(\binom{n}{k}\right).$$

For an admissible k, define its complementary carry weight by

$$W_n(k) = \prod_{p \mid n} p^{\max\{v_p(n) - C_p(n,k), 0\}}. \tag{1}$$

Fix a positive divisor q of n. Define

$$\mathsf{BoundaryAt}(n,q,d) \iff d \mid n, \quad q < d, \quad \frac{d}{r} \le q \text{ for every prime } r \mid d. \tag{2}$$

Thus d is a divisibility-minimal divisor of n lying numerically above q. Define

$$\mathsf{Realized}(n,d) \iff \begin{aligned} &\text{there is an } m \ge 1 \text{ such that } dm \le \lfloor n/2 \rfloor, \\ &C_r(n, dm) \le v_r(n) - v_r(d) \quad \text{for every prime } r \mid d. \end{aligned} \tag{3}$$

Finally,

$$\mathsf{BoundarySafeAt}(n,q) \iff \text{no } d \text{ satisfying (2) is realized.} \tag{4}$$

All ranges in (2)–(4) are finite, and the same multiplier m must satisfy every carry constraint in (3).

Theorem 1 (Prime-power threshold theorem)**.** *Let p be prime, let $b \geq 1$, and put $q = p^b$. If $q \mid n$ and $q < n$, then*

$$\boxed{f(n) = \frac{n}{q} \quad \Longleftrightarrow \quad \mathsf{BoundarySafeAt}(n, q).}$$

2 The exact complementary weight

Put $G_n(k) = \gcd(n, \binom{n}{k})$. Prime by prime,

$$v_p(G_n(k)) = \min(v_p(n), C_p(n,k)) .$$

Since $\min(a, c) + \max(a - c, 0) = a$, (1) gives the exact identity

$$\boxed{G_n(k)W_n(k) = n.} \tag{5}$$

In particular, $W_n(k)$ is a positive divisor of n.

The identity

$$n\binom{n-1}{k-1} = k\binom{n}{k}$$

shows that $n \mid k\binom{n}{k}$. Taking valuations gives

$$v_p(n) - C_p(n,k) \leq v_p(k) \quad \text{whenever the left side is positive,}$$

and hence

$$\boxed{W_n(k) \mid k.} \tag{6}$$

This divisibility is what preserves one common index across all primes.

3 Every proper prime-power threshold is attained

Write $n = cq$. Since $q < n$, we have $c \geq 2$, so $2 \leq q \leq \lfloor n/2 \rfloor$ and q is admissible. The binomial identity

$$\binom{cq}{q} = c\binom{cq-1}{q-1} \tag{7}$$

will give the exact value at this row. In base p, the lowest b digits of both $cq-1$ and $q-1 = p^b-1$ are $p-1$. Repeated application of Lucas's theorem therefore gives

$$\binom{cq-1}{q-1} \equiv 1 \pmod{p}. \tag{8}$$

If $A = \binom{cq-1}{q-1}$, then A is coprime to q, even when $p \mid c$. Consequently

$$\gcd\left(cq, \binom{cq}{q}\right) = \gcd(cq, cA) = c\gcd(q, A) = c = \frac{n}{q}. \tag{9}$$

Combining (5) and (9) yields

$$W_n(q) = q. \tag{10}$$

It follows that

$$f(n) = \frac{n}{q} \quad \Longleftrightarrow \quad W_n(k) \leq q \quad \text{for every admissible } k. \tag{11}$$

Indeed, (5) reverses the numerical order between $G_n(k)$ and $W_n(k)$, while (9)–(10) supply equality at an admissible row.

4 The boundary antichain

Lemma 2. *If $W \mid n$, then*

$$q < W \quad \Longleftrightarrow \quad \textit{some } d \mid W \textit{ satisfies } \mathsf{BoundaryAt}(n, q, d).$$

Proof. If $W > q$, choose the numerically least divisor d of W with $d > q$. Then $d \mid n$. For every prime $r \mid d$, the proper divisor d/r also divides W, so minimality gives $d/r \le q$. Thus d satisfies (2). The reverse implication follows from $q < d \le W$. ☐

For $d \mid n$, valuation comparison in (1) gives

$$d \mid W_n(k) \quad \Longleftrightarrow \quad C_r(n, k) \le v_r(n) - v_r(d) \quad \text{for every prime } r \mid d. \tag{12}$$

If the left side holds, (6) gives $d \mid k$, say $k = dm$ with $m \ge 1$. The admissible endpoint and (12) say exactly that d is realized. Conversely, a witness in (3), together with (12), gives $d \mid W_n(dm)$. Therefore

$$\begin{aligned} W_n(k) \le q \quad & \text{for every admissible } k \\ & \Longleftrightarrow \text{no boundary divisor above } q \text{ is realized} \\ & \Longleftrightarrow \mathsf{BoundarySafeAt}(n, q). \end{aligned} \tag{13}$$

Equations (11) and (13) prove Theorem 1.

5 The compact synchronized compiler

The preceding theorem is already an exact iff criterion, but its semantic definition quantifies over a shared multiplier. We now describe a compact form that removes any need to list those multipliers in the statement while retaining one symbolic shared multiplier.

Fix an ordered exact prime factorization

$$n = \prod_{i=1}^{r} p_i^{a_i}$$

and a baseline B with $2 \le B \le n$. A semantic factor tableau selects exponents $0 \le e_i \le a_i$ and one positive integer T. Put

$$d = \prod_{i=1}^{r} p_i^{e_i}, \qquad K = dT.$$

The tableau is feasible when

$$B < d, \qquad K \le \lfloor n/2 \rfloor,$$

and, for every active index i with $e_i > 0$,

$$\frac{d}{p_i} \le B, \qquad C_{p_i}(n, K) \le a_i - e_i. \tag{14}$$

Thus d is a boundary divisor and the one literal row $K = dT$ realizes it. Conversely every realized boundary divisor determines exactly such an exponent vector and shared T. Hence

$$\mathsf{FactorTableau}(n, B) \text{ is feasible} \quad \Longleftrightarrow \quad \exists d \, [\mathsf{BoundaryAt}(n, B, d) \wedge \mathsf{Realized}(n, d)]. \tag{15}$$

The explicit system $\mathcal{G}(F, B)$ is a natural-linear presentation of this tableau. It contains the following finite fields.

1. One-hot Boolean selectors choose each exponent $e_i \in \{0, \ldots, a_i\}$.

2. Prefix variables build d and $K = dT$ by multiplying only by the fixed coefficients p_i^e selected in the preceding step.
3. For every i, digit variables expand the same integer K in base p_i through height $h_i = \lfloor \log_{p_i} n \rfloor$.
4. Boolean borrow variables encode subtraction of K from n in each base. Their sum is exactly $C_{p_i}(n, K)$.
5. Boundary and budget rows enforce (14) when $e_i > 0$ and become inactive through checked big-M bounds when $e_i = 0$.

Every multiplication in these rows is by input data or a selected fixed prime power; there is no product of two free integer variables. The system is therefore a finite Presburger, equivalently mixed integer/Boolean, feasibility instance.

Theorem 3 (Compiler soundness and completeness)**.** *For every ordered exact factorization F of n and every $2 \le B \le n$,*

$$\boxed{\mathcal{G}(F, B) \iff \mathsf{FactorTableau}(n, B) \text{ is feasible.}}$$

Consequently, for composite $n > 1$,

$$\boxed{f(n) = \frac{n}{P(n)} \iff \neg\mathcal{G}(F, P(n)).} \tag{16}$$

Proof. From a satisfying assignment, one-hotness recovers the exponent vector. The active prefix equalities recover d and $K = dT$ exactly. The digit and borrow rows recover the genuine base-p_i subtraction, so the budget rows give the carry inequalities in (14). This constructs a semantic tableau.

Conversely, from a semantic tableau choose the corresponding one-hot selectors, use the actual prefix products of d and K, take the canonical base-p_i digits of K, and take the genuine borrows in $n - K$. These values satisfy every selector, bound, prefix, boundary, digit, borrow, and budget row. This proves the displayed equivalence. Equation (15), followed by Theorem 1 at $q = P(n)$, gives (16). □

This compilation is materially smaller than enumerating the possible multipliers. The indexed selector, digit, borrow, and prefix coordinates are linear in

$$\sum_i (a_i + 1) + \sum_i (h_i + 1) + r.$$

Since r, $\sum_i a_i$, and every h_i are bounded by the binary length of n, there are $O((\log n)^2)$ indexed coordinates. The fixed coefficients have $O(\log n)$ bits, giving an $O((\log n)^3)$ direct sparse description. This is a compact symbolic feasibility criterion, not a claim that the resulting feasibility problem is decidable in polynomial time. In particular, it does not eliminate the shared multiplier and is not a direct taxonomy of the equality cases from their prime factorizations.

6 The two formulations of Part (i)

Let $P(n)$ be the largest prime divisor of n.

Corollary 4 (Maintained largest-prime formulation)**.** *For every composite $n > 1$ and every ordered exact factorization F of n,*

$$\begin{aligned} f(n) = \frac{n}{P(n)} &\iff \mathsf{BoundarySafeAt}(n, P(n)) \\ &\iff \neg\mathcal{G}(F, P(n)). \end{aligned}$$

Proof. Apply Theorem 1 with $q = P(n) = P(n)^1$. Since n is composite, $P(n)$ is a proper divisor of n. The second equivalence is Theorem 3. $\square$

The 1978 paper instead uses

$$Q(n) = \max_{p|n} p^{v_p(n)}, \tag{17}$$

the greatest exact prime-power component of n.

Corollary 5 (Literal 1978 formulation)**.** *For every composite $n > 1$ and every ordered exact factorization F of n,*

$$\begin{aligned} f(n) = \frac{n}{Q(n)} \quad &\iff \quad n \textit{ is not a prime power and } \mathsf{BoundarySafeAt}(n, Q(n)) \\ &\iff \quad n \textit{ is not a prime power and } \neg\mathcal{G}(F, Q(n)). \end{aligned}$$

Proof. If n is not a prime power, an exact component attaining (17) is a proper prime-power divisor, so Theorem 1 applies. If $n = p^a$ is composite, then $Q(n) = n$. Lucas's theorem shows that every interior binomial coefficient is divisible by p, while Kummer's theorem at $k = p^{a-1}$ gives valuation exactly one. Thus $f(p^a) = p > 1 = n/Q(n)$, so the equality is impossible. The second equivalence follows from Theorem 3 at baseline $Q(n)$. $\square$

Remark 6. The two formulations are genuinely different. For $n = 12$, the gcd values for $k = 2, 3, 4, 5, 6$ are $6, 4, 3, 12, 12$, so $f(12) = 3$. Here $P(12) = 3$ and $Q(12) = 4$: the historical equality $f(12) = 12/Q(12)$ holds, whereas the maintained equality $f(12) = 12/P(12)$ does not.

7 Classification boundary

The displayed equivalences are exact, finite, and kernel-checked. They prove that the original equality question is equivalent to infeasibility of the explicit synchronized system. A stricter reading of "characterize", however, asks for conditions read directly from the ordered exact factors without solving this custom feasibility problem. The compiler does not meet that stronger bar: its one shared multiplier must still satisfy coupled digit and borrow conditions in every active prime base. A later audit reduces the remaining task to elimination of a short-interval CRT digit-cylinder intersection and records counterexamples to independent-prime, greedy, pairwise, and divisor-descent shortcuts. No exhaustive non-search branch theorem is currently proved. Thus this document establishes the exact finite criterion, not closure of that stricter direct-classification problem.

8 Formal verification and source trail

The parameterized theorem is `Erdos700PartI.f_eq_div_primePow_iff_boundarySafeAt`.
The maintained and historical corollaries are respectively
`Erdos700PartI.f_eq_div_iff_boundarySafe` and
`Erdos700PartI.erdos_700_i_historical`. They are exposed in
`lean-part1/PartIVerify.lean`. The repository verifier builds the promoted roots, rejects placeholders, audits dependencies, and runs finite regressions. Lean reports only the standard Mathlib axioms `propext`, `Classical.choice`, and `Quot.sound`; it reports no `sorryAx` dependency.
The semantic bridge is
`Erdos700PartI.factorTableauFeasible_iff_exists_boundary_realized`, and the explicit compiler theorem is `Erdos700PartI.ExplicitG.explicitG_iff_factorTableauFeasible`.
Lean checks both directions, including one-hot decoding, prefix products, canonical digit and borrow construction, inactive-row bounds, and the exact carry-budget identity.

The primary historical source is P. Erdős and G. Szekeres, “Some number theoretic problems on binomial coefficients,” *Australian Mathematical Society Gazette* 5 (1978), 97–99, archival PDF.